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Pneu-ConViT: A CLAHE-Enhanced Hybrid CNN-Vision Transformer Ensemble for Pneumonia Classification

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ABSTRACT

Pneumonia detection from chest X-ray (CXR) images remains a challenging task due to subtle radiographic patterns and large variations in disease presentation. Recently, vision transformers (ViTs) have demonstrated strong capability in modelling global contextual relationships, while convolutional neural networks (CNNs) remain highly effective in extracting local spatial features. In this study, a hybrid framework named Pneu-ConViT is proposed for automated pneumonia classification using chest radiographs. To enhance the visibility of subtle pulmonary abnormalities, contrast limited adaptive histogram equalisation is applied as a preprocessing step to improve local contrast in CXR images. The proposed model integrates multiple pre-trained architectures, including ViT-B/16, VGG16, DenseNet201, ResNet50V2, and EfficientNetB3, within a weighted late-fusion ensemble strategy. This design enables the combination of global self-attention representations from transformer models with hierarchical convolutional feature extraction from CNNs. Experiments conducted on the Kermany CXR dataset demonstrate that the proposed framework achieves an accuracy of 98.63%, an area under the ROC curve (AUC) of 99.90%, and an F1-score of 99.08%, outperforming several individual backbone models and recent deep learning approaches. To further assess generalisation capability, the model was evaluated on the RSNA Pneumonia Detection Challenge dataset, where it achieved an accuracy of 84.50%, an AUC of 87.88%, and an F1-score of 86.74%. Furthermore, model interpretability is investigated using gradient-weighted class activation mapping and transformer attention visualisation, revealing that the proposed system focuses on clinically meaningful lung regions associated with pneumonia. These results indicate that the proposed hybrid CNN-ViT framework provides an effective and interpretable solution for computer-aided pneumonia diagnosis from chest radiographs.

1 | Introduction

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, responsible for over 2.5 million deaths annually, particularly among children under five and elderly populations [1, 2]. It manifests as inflammation of the alveoli, leading to fluid accumulation that impedes gas exchange and oxygen absorption. Early detection is vital for improving survival rates and reducing complications; however, manual interpretation of chest X-ray (CXR) images is both time-consuming and highly dependent on radiologist expertise [3]. Variability among

clinicians and limited diagnostic resources in developing regions often delay treatment and reduce diagnostic accuracy [4, 5].

The advent of artificial intelligence (AI) and deep learning (DL) techniques has enabled the development of computer-aided diagnosis (CAD) systems that provide consistent and efficient medical image analysis [6]. DL methods, particularly convolutional neural networks (CNNs), have demonstrated strong performance across a wide range of computer vision and biomedical applications [7, 8]. CNN-based architectures such as VGG16 [9], ResNet50 [10], DenseNet201 [11], and EfficientNetB3 [12] have achieved

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high accuracy in pneumonia classification tasks, owing to their capability to extract hierarchical and discriminative features from medical images.

Nevertheless, CNNs exhibit certain limitations due to their localised receptive fields, which constrain their ability to model global contextual dependencies. As a result, CNN-based models may overlook spatially distant yet clinically correlated lung patterns, particularly in early-stage or diffuse pneumonia cases [13]. To overcome these constraints, recent research has focused on the Vision Transformer (ViT) architecture [14], which replaces convolutional operations with self-attention mechanisms, allowing the model to learn long-range dependencies and holistic spatial relationships across the entire image [15, 16].

However, ViTs often require large-scale annotated datasets for optimal training and tend to underperform on limited medical datasets [17]. Additionally, ViTs may fail to capture fine-grained local details that CNNs inherently learn through convolutional feature extraction [18]. This has motivated research into hybrid CNN-ViT architectures, which aim to leverage the complementary strengths of both paradigms. CNNs are proficient in capturing low-level spatial patterns, while Transformers excel at modelling global dependencies [19].

Hybrid frameworks such as Swin Transformer [20] and Swin-Unet [21] have achieved promising results in medical imaging classification and segmentation tasks. However, most prior studies integrate only one CNN backbone with a Transformer encoder and are limited in their ability to fuse multiple diverse models or generalise across datasets. There remains a gap in designing a comprehensive hybrid ensemble that effectively combines multiple CNN and Transformer architectures for robust, generalisable pneumonia detection.

To address these challenges, this paper proposes a hybrid ViT-CNN ensemble model that combines five powerful pre-trained architectures – ViT-B/16, VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 – using a weighted late-fusion ensemble strategy. Each model contributes distinct yet complementary feature representations, improving both robustness and interpretability. The proposed framework is evaluated on the Kermany Pneumonia dataset [22], a widely used benchmark for CXR image classification. This dataset contains labelled paediatric chest radiographs categorised into Pneumonia and Normal classes, providing a reliable foundation for assessing model performance in medical image classification. The images were collected under varying clinical conditions and imaging protocols, offering natural variability in brightness, contrast, and anatomical presentation. Such diversity enables a robust evaluation of the proposed Pneu-ConViT framework's generalisation capability, diagnostic reliability, and resilience to real-world imaging noise.

The main contributions of this study are summarised as follows:

- We propose Pneu-ConViT, a hybrid CNN-Transformer framework that integrates ViT-B/16 with multiple complementary CNN backbones, including VGG16, DenseNet201, ResNet50V2, and EfficientNetB3, to leverage both local and global feature representations for pneumonia classification.
- A learnable deep neural network (DNN)-based feature fusion module is introduced to effectively combine heterogeneous CNN and Transformer features at the representation level, enabling more informative and discriminative feature learning than conventional decision-level fusion approaches.
- The framework incorporates contrast-limited adaptive histogram equalisation (CLAHE)-based contrast enhancement and generative adversarial network (GAN)-based data augmentation to address common challenges in CXR analysis, including image quality variations, data scarcity, and class imbalance.
- Extensive experiments on the Kermany Pneumonia dataset, together with five-fold cross-validation and external validation on the RSNA dataset, demonstrate the robustness, generalisation capability, and effectiveness of the proposed framework.
- Gradient-weighted class activation mapping (Grad-CAM) and Transformer attention visualisations are employed to highlight clinically relevant regions, improving model interpretability and supporting transparent AI-assisted diagnosis.
- The novelty of Pneu-ConViT lies not in any individual component but in the integration of multi-backbone CNN feature extraction, ViT-based global contextual modelling, and learnable feature-level fusion within a unified end-to-end framework for pneumonia classification.

The remainder of this paper is organised as follows. Section 2 reviews prior work on CNN-based and Transformer-based methods for pneumonia detection. Section 3 details the datasets and proposed hybrid ensemble methodology. Section 4 presents experimental results and comparative analysis, while Section 5 concludes the study and discusses future research directions.

2 | Related Works

The development of DL-based CAD systems for pneumonia classification has evolved rapidly in recent years. This section reviews three major directions in the literature: (1) CNN-based methods, (2) Transformer-based vision models, and (3) hybrid and ensemble learning strategies integrating both paradigms.

2.1 | CNN-Based Pneumonia Detection

CNNs have long been the foundation of medical image classification because of their powerful ability to learn spatial hierarchies through local receptive fields. One of the earliest DL studies on pneumonia detection, by Jiang et al. [23], applied transfer learning using VGG16 to classify normal versus pneumonia CXRs. This work demonstrated that features pre-trained on natural images could effectively generalise to medical imaging tasks.

Following that, Bunde et al. [24] introduced DenseNet architectures, such as DenseNet121, DenseNet169, and DenseNet201 trained on a pneumonia dataset, which achieved radiologist-level accuracy in pneumonia detection. Similarly, the ResNet50V2 model [25] later improved residual connections and optimisation stability, facilitating deeper feature extraction. Recent efforts

focused on efficiency and scalability, exemplified by EfficientNetB0 [26], which uses compound scaling to balance network depth, width, and resolution. Its success in both medical and non-medical domains has made it a popular backbone for transfer learning-based pneumonia detection [27]. Studies combining CNNs with advanced augmentation, such as rotation, CLAHE filtering, and contrast stretching, reported improved robustness across datasets [28, 29].

Despite these improvements, CNNs remain limited by their inherently local receptive fields, which constrain their ability to model long-range dependencies within an image [30]. This makes it difficult for CNNs to capture contextual relationships between distant lung regions, which can be crucial for differentiating subtle pneumonia manifestations from healthy structures.

2.2 | Vision Transformers in Medical Imaging

The introduction of ViTs [14] marked a paradigm shift in computer vision, replacing convolutions with self-attention mechanisms that capture global spatial dependencies. ViTs divide an image into non-overlapping patches, treat them as sequential tokens, and process them using multi-head self-attention (MHSA) layers to model long-distance relationships effectively. This approach has been particularly attractive for medical imaging, where capturing both global and local structures is critical [31].

In the medical domain, ViT-based models have demonstrated superior interpretability and adaptability. For example, TransMed [32] introduced a Transformer architecture for multi-modal medical classification. Pham et al. [33] demonstrated that ViTs outperform CNNs in pneumonia classification by modelling broader contextual cues, while Marikkar et al. [34] designed a ViT architecture to perform multi-label classification on CXR images, effectively capturing long-range dependencies between lung regions to detect multiple co-occurring diseases with high F1-score and AURO. However, ViTs require large annotated datasets and are prone to overfitting when trained on limited medical data [35, 36]. Additionally, they often underperform in capturing local texture variations that CNNs naturally learn, highlighting the potential benefit of hybrid approaches.

2.3 | Hybrid and Ensemble Architectures

To address the complementary limitations of CNNs and ViTs, researchers have explored hybrid architectures that combine the local feature learning of CNNs with the global context modelling of Transformers. The seminal ConViT model [37] incorporated convolutional inductive biases into self-attention layers, improving data efficiency. Similarly, AddViT [38] introduced architectural refinements to build deeper, more stable hybrid networks. In the medical field, the Swin Transformer [20] has become a foundational component for hybrid medical image models due to its efficiency and strong representation capability across scales. These hybrids have shown consistent improvements over pure CNN or ViT baselines in segmentation, classification, and detection tasks [39, 40].

Another complementary direction is ensemble learning, where multiple independent networks are combined to reduce variance and enhance generalisation [41]. Ensembles have proven particularly beneficial in medical imaging, where data variability and noise can severely impact model reliability. For example, an ensemble integrating VGG19, ResNet50V2, and DenseNet121 backbones has achieved strong performance in chest disease classification [42]. Similarly, another study compares three transfer learning models – DenseNet169, VGG16, and InceptionV3 – and applies ensemble strategies to produce more accurate pneumonia predictions [43].

However, most prior works either rely on a single hybrid architecture or ensemble models limited to CNNs. Very few have investigated multi-model hybrid ensembles that integrate both CNN and Transformer backbones across multiple datasets, particularly for cross-domain pneumonia detection. This motivates our proposed Hybrid ViT–CNN Ensemble Framework, which jointly leverages ViT-B/16, VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 to capture both local and global representations while maintaining robust generalisation across the Kermany dataset.

3 | Methodology

This section details the overall pipeline of the Pneu-ConViT framework for pneumonia classification, including dataset descriptions, preprocessing strategies, model architecture, and the ensemble fusion mechanism.

3.1 | Dataset

To ensure a robust and clinically meaningful evaluation, experiments were conducted on the Kermany Pneumonia dataset [22] and the RSNA dataset [44], the publicly available and widely adopted benchmark for CXR image classification. These datasets provide a diverse and representative collection of paediatric chest radiographs, enabling a comprehensive assessment of the Pneu-ConViT framework's diagnostic accuracy, generalisation capability, and interpretability in pneumonia detection.

3.1.1 | Kermany Pneumonia Dataset

The dataset curated by Kermany et al. [22] consists of paediatric chest radiographs collected from Guangzhou Women and Children's Medical Center. It contains 5856 images, categorised into Normal (1583) and Pneumonia (4273) cases, with the pneumonia class further subdivided into bacterial and viral subtypes. The dataset's partitioning corresponds approximately to training (75%), validation (15%), and testing (10%) subsets using a patient-wise stratified approach to prevent data leakage, illustrated in Table 1. All images are greyscale with varying resolutions, typically around 1024×1024 pixels.

3.1.2 | RSNA Pneumonia Dataset

The RSNA Pneumonia Detection Challenge dataset [44] consists of approximately 30,000 CXR images collected from the

TABLE 1 | The number of samples trained and tested after splitting.

Class	Training set	Validation set	Testing set
Normal	1188	237	158
Pneumonia	3204	641	428
Total	4392	878	586

TABLE 2 | Augmentation parameters.

Class	Training set
Rescale	1/255
Shear_range	10
Zoom_range	0.2
Horizontal_flip	True
Brightness_range	(0.9,1.1)
Width_shift_range	0.1
Rotation_range	15

NIH ChestX-ray14 database and further annotated by expert radiologists. The dataset is officially divided into 26,684 training images and 3000 test images, while no predefined validation set is provided. Consequently, a portion of the training data is typically reserved for validation during model development to ensure reliable performance assessment. Each image is categorised into Normal, No Lung Opacity/Not Normal, or Lung Opacity (Pneumonia), with pneumonia cases additionally annotated using bounding boxes that indicate regions of lung opacity. This detailed annotation enables the dataset to support both classification and localisation tasks, making it widely used for developing and evaluating DL models in pneumonia detection.

3.2 | Preprocessing via CLAHE and Data Augmentation

Proper preprocessing is crucial for stabilising training and improving generalisation, particularly for medical images that vary widely in contrast and illumination.

- **Normalisation:** All images were normalised to the [0, 1] range after pixel-intensity scaling.
- **Resizing:** Images were resized to 224×224 pixels, consistent with ViT-B/16 and CNN backbone input requirements.
- **Augmentation:** Random transformations were applied, including rotation ($\pm 15^\circ$), horizontal flipping, brightness shift ($\pm 10\%$), and zoom ($\pm 20\%$). This strategy mitigated overfitting while enhancing robustness to anatomical and acquisition variations, illustrated in Table 2.
- **CXR images often suffer from low contrast and uneven illumination, which may obscure subtle pneumonia-related abnormalities. To enhance local contrast and improve the visibility of pulmonary structures, CLAHE [45] is applied as a preprocessing step. CLAHE divides the input image into small contextual regions (tiles) and performs histogram**

equalisation independently within each region. The enhanced pixel intensity is obtained from the cumulative distribution function (CDF) of the local histogram (Equation (1))

$$I'(x, y) = (L - 1) \sum_{k=0}^{I(x,y)} \frac{h(k)}{MN} \quad (1)$$

where $I(x, y)$ is the original pixel value, $h(k)$ denotes the histogram of grey level k , $M \times N$ is the number of pixels in the tile, and L represents the number of grey levels. To prevent excessive noise amplification, CLAHE introduces a **clip limit** that constrains the maximum histogram value (Equation (2))

$$T = \frac{\text{clip_limit} \times M \times N}{L} \quad (2)$$

where $M \times N$ denotes the number of pixels in each tile and L is the number of grey levels. The clipped histogram is redistributed and converted into a CDF to obtain the enhanced pixel intensity. Bilinear interpolation is then used to remove boundary artefacts between neighbouring tiles. The images are subsequently resized to 224×224 pixels and normalised prior to being fed into the DL models. This preprocessing step enhances local lung contrast and improves the ability of the proposed hybrid ViT-CNN model to extract diagnostically relevant features.

3.3 | The Hybrid ViT-CNN Architecture

The workflow is illustrated conceptually in Figure 1. The hybrid ensemble model integrates five pre-trained architectures – ViT-B/16, VGG16, ResNet50V2, DenseNet201, and EfficientNetB3 – each serving as a base learner for complementary feature extraction. The feature embeddings from all networks are subsequently fused to form a unified diagnostic representation. Finally, the concatenated feature vector is passed through a DNN that performs prediction.

3.3.1 | Vision Transformer Backbone

The ViT-B/16 splits each input image into 14×14 patches, each measuring 16×16 pixels. These patches are linearly projected into embeddings and combined with a learnable classification token. Positional encodings are incorporated to maintain spatial relationships, followed by several MHSA and feed-forward layers. This architecture allows the ViT to capture global context, effectively distinguishing pneumonia-affected areas that might be far apart spatially but linked through pathology.

3.3.2 | Convolutional Feature Extractors

The CNN backbones contribute localised structural representations: VGG16 provides deep hierarchical feature maps through sequential convolution and max-pooling operations. ResNet50V2 employs identity shortcut connections, mitigating vanishing gradients and enabling deeper learning. DenseNet201 facilitates dense inter-layer connectivity to maximise feature reuse and strengthen gradient flow. EfficientNetB3 balances network depth,

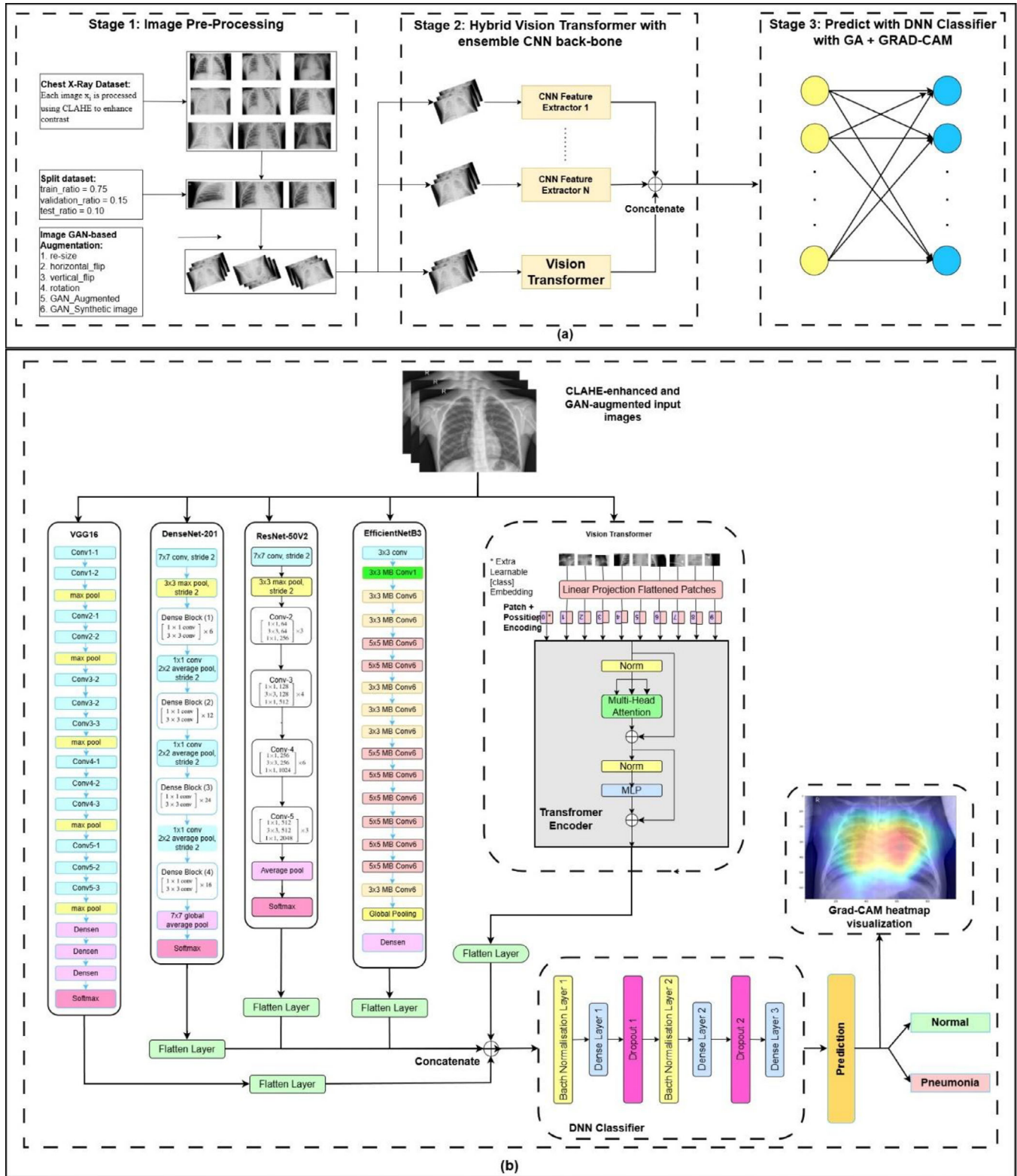


FIGURE 1 | Illustration of the overall strategy for the Pneu-ConViT mode, a hybrid approach that combines the vision transformer with effective CNN backbone models. (a) Our Pneu-ConViT framework consists of three main stages: Stage 1 involves pre-processing, Stage 2 utilises an ensemble of effective CNN models and vision transformer, and Stage 3 employs a DNN classifier to achieve the best results. (b) The details flow from the input X-ray image to the ensemble of four CNN models (VGG16, DenseNet-201, ResNet50-V2, and EfficientNetB3) and the vision transformer, then to the DNN classifier to predict normal or pneumonia.

width, and resolution using compound scaling for high accuracy-to-parameter efficiency. Each CNN backbone was initialised with ImageNet pre-trained weights and fine-tuned on the pneumonia datasets using transfer learning. The final convolutional feature vectors were extracted from the penultimate fully connected layer before the classification head.

3.3.3 | Feature Fusion and Representation Alignment

To ensure consistency across models, all feature vectors were projected into a common 1024-dimensional latent space using a learnable fully connected projection. Layer normalisation and dropout (0.3) were applied to mitigate overfitting. The projected features were then concatenated with ReLU activations, producing the ensemble's fused embedding vector F_i (Equation (3)).

$$F_i = \text{Concat}(f_{\text{ViT}}, f_{\text{VGG}}, f_{\text{RESNET}}, f_{\text{DENSE}}, f_{\text{EFF}}) \quad (3)$$

3.3.4 | Classification Head (DNN-Based Fusion Layer)

The concatenated feature vector F_i is passed through a DNN-based learnable fusion module that performs feature compression, non-linear transformation, and final decision-making. The DNN-based learnable fusion module layer serves as the final classification head following the multimodal feature concatenation of the ViT-B/16 and CNN branches. It consists of three fully connected layers with progressively decreasing neuron sizes – Dense (512, ReLU), Dense (256, ReLU), and Dense (128, ReLU) – each followed by a Dropout layer with a rate of 0.5 to mitigate overfitting and enhance model generalisation. The ReLU activation function introduces nonlinearity, enabling the network to learn complex feature interactions among the fused representations. The high-dimensional Dense (512) layer captures rich semantic correlations, while the subsequent Dense (256) and Dense (128) layers refine and compress the feature space into more discriminative embeddings. After the DNN fusion block, the final layer applies a Dense layer with Sigmoid activation to produce the binary output for pneumonia detection. This activation maps the output of the network to a probability value between 0 and 1, representing the likelihood that a given CXR corresponds to pneumonia. This hierarchical fusion structure effectively balances model expressiveness and regularisation, ensuring stable and accurate predictions in pneumonia image classification.

3.4 | Ensemble Fusion Mechanism

While deep hybrid architectures can learn rich representations, single-model predictions may still exhibit uncertainty, especially when visual symptoms are subtle. To address this, a weighted late-fusion ensemble was designed, integrating probabilistic outputs from each backbone. The optimisation framework is composed of two sequential stages. First, genetic algorithm (GA) is employed during the training phase to optimise model-level hyperparameters, including learning rate, weight decay, batch size, dropout rate, and the number of unfrozen layers. This process determines the optimal configuration for each individual CNN and ViT model. After training is completed, feature-level

fusion is performed through a DNN-based fusion module. In this stage, the feature representations extracted from the trained backbone models are concatenated and passed through an MLP to learn the optimal nonlinear combination of complementary features. Unlike traditional weighted ensemble methods, the fusion network is trained jointly to automatically learn discriminative feature interactions without requiring manually predefined or optimised fusion weights. The test set remains completely unseen and is used only for final evaluation.

3.4.1 | GA-Based Hyperparameter Optimisation for Overfitting Mitigation

To mitigate the risk of overfitting, a GA-based hyperparameter optimisation strategy is incorporated into the training pipeline. While GA is not a direct regularisation method, it serves as an adaptive search mechanism that identifies hyperparameter configurations with better generalisation performance by optimising validation metrics rather than training accuracy.

In this framework, each chromosome encodes a set of key hyperparameters, including learning rate, dropout rate, weight decay, batch size, and the number of unfrozen layers. Candidate solutions are evaluated using a fitness function defined on validation performance (e.g., area under the ROC curve (AUC) or F1-score), ensuring that selected configurations generalise well to unseen data. The evolutionary process applies selection, crossover, and mutation to iteratively refine the population and explore the hyperparameter space. By emphasising hyperparameters related to model capacity and regularisation, such as dropout and weight decay, GA helps identify configurations that balance model complexity and generalisation ability. Compared to manual tuning, this approach provides a more efficient and robust search strategy in Figure 2.

Overall, GA contributes to overfitting mitigation indirectly by enabling validation-driven hyperparameter selection, thereby improving the robustness and generalisation of the proposed hybrid CNN-ViT model.

3.4.2 | DNN-Based Fusion

Each backbone model M_i extracts a feature vector from the input CXR image. The feature vectors generated by ViT, VGG16, ResNet50V2, DenseNet201, and EfficientNetB3 are concatenated to form a unified feature representation (Equation (4)).

$$F_{\text{Concat}} = \text{Concat}(f_{\text{ViT}}, f_{\text{VGG}}, f_{\text{RESNET}}, f_{\text{DENSE}}, f_{\text{EFF}}) \quad (4)$$

where $(f_{\text{ViT}}, f_{\text{VGG}}, f_{\text{RESNET}}, f_{\text{DENSE}}, \text{ and } f_{\text{EFF}})$ denote the feature vectors extracted from the corresponding backbone networks.

The concatenated feature vector is then processed by a DNN-based fusion network consisting of multiple fully connected layers and dropout regularisation (Equation (5)).

$$F_i = \text{MLP}(F_{\text{Concat}}) \quad (5)$$

Generation 3/5	
1. AUC=0.9990	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.1, 'unfreeze_layers': 2}
2. AUC=0.9989	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
3. AUC=0.9986	{'lr': 0.0001, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 1}
4. AUC=0.9985	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.5, 'unfreeze_layers': 2}
5. AUC=0.9982	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.1, 'unfreeze_layers': 2}
6. AUC=0.9979	{'lr': 5e-05, 'weight_decay': 0.001, 'batch_size': 32, 'dropout': 0.1, 'unfreeze_layers': 2}

Generation 4/5	
1. AUC=0.9991	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
2. AUC=0.9991	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.5, 'unfreeze_layers': 2}
3. AUC=0.9990	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.1, 'unfreeze_layers': 2}
4. AUC=0.9982	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
5. AUC=0.9974	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 1}
6. AUC=0.9907	{'lr': 1e-05, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.5, 'unfreeze_layers': 2}

Generation 5/5	
1. AUC=0.9991	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 32, 'dropout': 0.5, 'unfreeze_layers': 2}
2. AUC=0.9988	{'lr': 0.0005, 'weight_decay': 1e-05, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
3. AUC=0.9985	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.1, 'unfreeze_layers': 2}
4. AUC=0.9975	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
5. AUC=0.9969	{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}
6. AUC=0.9957	{'lr': 0.0005, 'weight_decay': 0.001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}

Best chromosome found:	
{'lr': 0.0005, 'weight_decay': 0.0001, 'batch_size': 16, 'dropout': 0.5, 'unfreeze_layers': 2}	
Best validation AUC: 0.9991	

FIGURE 2 | Best AUC and parameters after GA optimisation.

Finally, the fused feature representation is passed through a sigmoid classification layer to obtain the probability of the pneumonia class (Equation (6)).

$$P_{\text{final}} = \sigma(F_i) \quad (6)$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function and P_{final} represents the final predicted probability.

The fusion network is trained using the training dataset and optimised through backpropagation together with the classification objective. The validation set is used solely for model selection and hyperparameter tuning, while the test set remains completely unseen until the final evaluation stage to ensure unbiased performance assessment.

3.4.3 | Final Classification and Prediction

Following feature fusion, the DNN-based fusion module generated the final classification probability through a sigmoid activation function. Class labels were assigned using a standard threshold of 0.5, where probabilities greater than or equal to 0.5 were classified as pneumonia and those below 0.5 were classified as normal.

3.4.4 | Explainability Integration

To enhance clinical interpretability, the ensemble integrates Grad-CAM [46]. Saliency maps from each CNN and attention maps from ViT are combined through attention-weighted averaging, yielding interpretable visual explanations of regions influencing model predictions.

3.5 | Mathematical Representation of the Pneu-ConViT Framework

3.5.1 | Algorithm of the Hybrid Architecture

The proposed hybrid framework leverages the ensemble of five distinct and complementary pre-trained models. Each contributes unique representational strengths, enabling superior generalisation and feature extraction across complex X-ray image datasets. We present a mathematical formulation of the hybrid network and provide theoretical reasoning behind its improved performance. A description of this hybrid network formulation is presented in Algorithm 1, illustrated in Table 3.

3.5.2 | Attention From ViT + Local Inductive Bias of CNN

CNNs have strong inductive biases (translation invariance, local connectivity). Transformers, however, have weaker inductive biases but greater flexibility to learn global patterns through self-attention. Given the attention matrix in ViT as defined in Equation (17).

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (17)$$

where $Q, K \in \mathbb{R}^{n \times d_k}$ are the query and key matrices, and ViT learns global relationships among patches.

Combining these two leads to a hybrid model that benefits from both as defined in Equation (12)

- CNN: $f_{\text{CNN}}(I) \approx \text{Local features}$

TABLE 3 | Algorithm 1 – formulation of the hybrid ViT–CNN network.

Step	Description
Input:	Let the input X-ray image (chest X-ray dataset) be denoted in Equation (7). $D = x_i \in \mathbb{R}^{224 \times 224 \times 3} \quad (7)$
Output:	Predicted class $\hat{y} = \{\text{Normal, Pneumonia}\}$
Step 1	Pre-processing —Each chest X-ray image x_i is first processed using Contrast Limited Adaptive Histogram Equalisation (CLAHE) to enhance local contrast.
Step 2	Dataset Split —Divide D into training (75 %), validation (15 %), and test (10 %) sets.
Step 3	Model Initialisation —Load five ImageNet-pretrained backbones: • M_1 : ViT-B/16 • M_2 : VGG16 • M_3 : DenseNet201 • M_4 : ResNet50V2 • M_5 : EfficientNetB3.
Step 4	Freeze lower layers; fine-tune top k blocks on the Kermamy dataset to adapt domain-specific features.
Step 5	Feature Extraction —For each image x_i , compute feature maps: $f_j = M_j(x_i)$, ($j = 1 \dots 5$). The image is fed into five parallel encoders as defined in Equations (8–12). $f_{\text{ViT-B16}} = \text{ViT_B16}(x_i) \quad (8)$ $f_{\text{VGG16}} = \text{VGG16}(x_i) \quad (9)$ $f_{\text{DenseNet201}} = \text{DenseNet201}(x_i) \quad (10)$ $f_{\text{ResNet50V2}} = \text{ResNet50V2}(x_i) \quad (11)$ $f_{\text{EffNetB3}} = \text{EfficientNetB3}(x_i) \quad (12)$
Step 6	Feature Normalisation —Apply L2-normalisation on each feature vector as defined in Equation (13). $\hat{f}_j = \frac{f_j}{\ f_j\ _2} \quad (13)$
Step 7	Feature Fusion —Concatenate normalised features to form a single high-dimensional representation $F_i = [\hat{f}_1; \hat{f}_2; \hat{f}_3; \hat{f}_4; \hat{f}_5]$ as defined in Equation (14). $F = \text{Concat}(F_1, F_2, F_3, F_4, F_5) \in \mathbb{R}^d \quad (14)$ where $d = d_1 + d_2 + d_3 + d_4 + d_5$
Step 8	DNN-based Learnable Logit Fusion : Feed F_i to a fully connected DNN layer: Dense (512, ReLU) → Dropout (0.5) → Dense (256, ReLU) → Dropout (0.5) → Dense (128, ReLU) → Dropout (0.5).
Step 9	Prediction : The final layer of the DNN employs a Sigmoid activation to transform the linear output z into a probability value $\hat{y} \in (0,1)$, as defined in Equation (15). $\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (15)$ $z = W^T F + b$. Here, W denotes the weight matrix of the final layer, b is the bias term, and F represents the fused feature vector obtained from the preceding layers.
Step 10	Loss Function —Compute categorical cross-entropy, as defined in Equation (16). $\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(P(x_i)) \quad (16)$
Step 11	Optimisation —AdamW optimiser with GA-based hyperparameter Optimisation (learning rate = 5×10^{-4} , weight_decay = 1×10^{-4} , dropout rate = 0.5, and batch size = 16). Early stopping (patience = 10) was used to avoid overfitting.
Step 12	Evaluation —On test set, compute Accuracy, Precision, Recall, F1-score, and AUC.
Step 13	Explainability —Generate Grad-CAM heatmaps for each backbone and ensemble to visualise salient pulmonary regions.
Return:	Final predicted label $\hat{y}$, confidence score $P(x_i)$, quantitative performance metrics, and Grad-CAM visual explanation.

- ViT: $f_{\text{ViT}}(I) \approx \text{Global context}$

$$f_{\text{Hybrid}}(I) = \text{Concat}[f_{\text{CNN}}(I), f_{\text{ViT}}(I)]$$

Thus, the final representation is more expressive.

3.5.3 | Computational Considerations

The model's complexity can be approximated by its parameter count as defined in Equation (18).

$$\text{Total Parameters} = \sum_{i=1}^5 \theta_i + \theta_{\text{FC}} \quad (18)$$

where θ_i is the number of parameters in the i_{th} encoder, and θ_{FC} is in the classification head. While computationally heavier than single models, its inference time remains manageable due to parallel feature extraction.

3.6 | Implementation Details

All models were implemented in TensorFlow 2.15 and trained using the AdamW optimiser with GA-based hyperparameter optimisation (a learning rate of 5×10^{-4} , a weight decay of 1×10^{-4} , a batch size of 16, and a dropout rate of 0.5). Early stopping (patience = 10) was applied to prevent overfitting. The batch size was set to 16, and all experiments were conducted on an NVIDIA A100 GPU (40 GB). Fine-tuning was performed for 50 epochs, with lower convolutional layers frozen during the first 10 epochs.

For transfer learning, each CNN backbone (VGG16, DenseNet201, ResNet50V2, and EfficientNetB3) was partially fine-tuned, where approximately the top 20%–30% of layers were unfrozen while the remaining layers were kept frozen. For the ViT-B/16, only the classification head and the final transformer blocks were fine-tuned, while earlier layers were frozen to preserve pre-trained representations.

Deep features extracted from the individual CNN backbones and the ViT-B/16 model were concatenated and fed into a DNN-based fusion module. The fusion network was trained end-to-end to learn the optimal integration of complementary local and global representations, producing the final classification output.

To ensure reproducibility, all experiments were conducted with a fixed random seed (seed = 42) across NumPy, TensorFlow, and data-loading operations. The dataset was split using a stratified train–validation–test protocol to maintain class distribution, and all reported results were averaged over three independent runs to reduce variance and improve robustness.

3.7 | Evaluation Metrics

Performance was quantified using standard metrics: Accuracy (Acc), Precision (Pr), Recall (Re), F1-score (F1), and AUC. For binary classification (normal vs. pneumonia), these metrics are computed in Equations (19–22).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (19)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (20)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (21)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

3.8 | Summary

The proposed Pneu-ConViT framework integrates multiple CNN backbones with ViT-B/16 to exploit complementary local and global feature representations for pneumonia classification. Unlike many existing hybrid CNN–Transformer approaches that rely on decision-level fusion or weighted averaging, the proposed framework employs a learnable DNN-based feature fusion module to combine heterogeneous CNN and Transformer features at the representation level. This design enables the model to capture complex interactions between local texture patterns and global contextual information, resulting in more effective feature integration and improved classification performance. Extensive evaluation on the Kermany Pneumonia dataset, together with external validation on the RSNA dataset, demonstrates the robustness, generalisation capability, and clinical applicability of the proposed framework. Furthermore, the integration of Grad-CAM and Transformer attention visualisation enhances model interpretability, supporting more transparent and reliable AI-assisted pneumonia diagnosis.

4 | Experimental Results and Discussion

This section presents the experimental evaluation of the Pneu-ConViT framework on the Kermany pneumonia dataset. The results include quantitative comparisons with baseline models, qualitative interpretability analysis, and ablation studies to highlight the effectiveness of the fusion mechanism. All reported metrics represent the mean of three independent runs to ensure statistical reliability.

4.1 | Experimental Setup

Training was performed on an NVIDIA A100 GPU (40 GB) with TensorFlow 2.15. Each model was trained for 50 epochs, using binary cross-entropy loss, the AdamW optimiser with a learning rate of 5×10^{-4} , a weight decay of 1×10^{-4} , a batch size of 16, and a dropout rate of 0.5. Early stopping (patience = 10) was used to avoid overfitting. Model weights from the best validation F1-score were saved for final evaluation.

For fair comparison, all CNN and ViT baselines were fine-tuned under identical conditions. Their extracted features were concatenated and integrated through a DNN-based fusion module to generate the final prediction.

TABLE 4 | Performance comparison of individual CNN models and ViT-B/16.

Mode	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
VGG16	88.9	93.73	88.21	90.89
ResNet50V2	90.87	89.18	97.18	93.01
DenseNet201	91.19	92.58	92.82	92.70
EfficientNetB3	95.56	98.57	95.39	96.95
ViT-B/16	93.75	95.24	92.31	93.75
Pneu-ConViT	98.63	99.31	99.54	99.08

TABLE 5 | Performance comparison of ensemble modes.

Mode	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
DenseNet201, VGG16, ResNet50V2	93.27	94.85	94.36	94.60
VGG16, MobileNetV2, ResNet50V2	94.87	94.97	96.92	95.94
ViT-B/16	93.75	95.24	92.31	93.75
Pneu-ConViT without a DNN-based fusion module	90.96	94.40	93.32	93.86
ResNet50V2 and ViT-B/16	96.95	94.87	97.37	96.11
Pneu-ConViT without EfficientNetB3	98.46	98.85	99.07	98.96
Pneu-ConViT	98.63	99.31	99.54	99.08

4.2 | Quantitative Results of Individual CNN Models and ViT-B/16

The proposed hybrid framework achieved the best overall performance across all evaluated metrics (Table 4). Individual CNNs performed strongly, while ViT alone underperformed slightly due to the dataset's limited size. However, their fusion substantially improved both sensitivity and AUC, indicating that complementary local-global features enhanced decision consistency.

The hybrid ensemble improved the baseline DenseNet201 and EfficientNetB3 results by approximately +3% in accuracy and +2% in F1-score, confirming that the proposed DNN-based fusion, built upon feature concatenation, enhances the model's ability to capture complementary and non-linear interactions across CNN and ViT representations, thereby improving decision confidence and reducing the misclassification of visually ambiguous cases.

4.3 | Quantitative Results of Ensemble Modes

The performance comparison of different ensemble and hybrid configurations demonstrates clear trends in how architectural diversity and model integration influence classification outcomes. The results of Table 5 show that hybridising CNNs with Transformer-based architectures, and particularly the Pneu-ConViT framework, yields substantial improvements across all evaluation metrics. First, the CNN-only ensemble modes (DenseNet201 + VGG16 + ResNet50V2 and VGG16 + MobileNetV2 + ResNet50V2) reveal moderate variations in performance depending on the choice and redundancy of architectures.

Among them, the ensemble of VGG16, MobileNetV2, and ResNet50V2 achieved the best CNN-only results, with 94.87% accuracy and 95.94% F1-score, indicating that combining both lightweight (MobileNetV2) and deep (ResNet50V2, VGG16) networks promotes complementary feature extraction. Ensembles containing more architectural redundancy – such as two ResNet50V2 models – perform slightly lower, suggesting that diversity among base learners improves ensemble strength. The standalone ViT-B/16 Transformer underperforms substantially with 93.8% accuracy and 92% F1-score. This outcome aligns with known limitations of ViTs when trained on limited datasets: they lack the strong spatial inductive biases inherent in CNNs, making them more dependent on large-scale data for effective generalisation. The weak recall and precision further indicate difficulty in capturing local texture features that CNNs handle more naturally. However, when the ViT-B/16 is paired with a CNN – specifically ResNet50V2 and ViT-B/16 – performance significantly increases to 96.95% accuracy and 96.11% F1-score. This improvement highlights the complementary strengths of the two architectures: ResNet50V2 provides robust local feature extraction, while ViT-B/16 contributes global contextual understanding. The hybrid synergy enables the model to address the limitations of each architecture, resulting in more balanced class detection, as evidenced by the high recall (97.37%).

The Pneu-ConViT framework achieves the best metrics across the entire comparison, with 98.63% accuracy, 99.31% precision, 99.54% recall, and 99.08% F1-score. These near-saturated performance values underscore the effectiveness of the proposed integration strategy – likely due to optimised multimodal feature fusion, improved regularisation, and maximised architectural complementarity. The exceptionally high precision indicates strong discriminative ability with minimal false positives, while the high

TABLE 6 | The overall accuracy and the averages of precision, recall, and F1-score (RSNA dataset [44]).

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
DenseNet201, VGG16, ResNet50V2	74	80	71	75
ResNet50V2, DenseNet201, ResNet50V2	76	79	76	78
DenseNet201, MobileNetV2, ResNet50V2	78	81	79	80
ResNet50V2, MobileNetV2, VGG16	78	82	75	79
ResNet50V2, DenseNet201, VGG16	70	76	66	71
ResNet50V2 and ViT-B16	84	84	86	85
Pneu-ConvViT	84.45	85.05	88.75	86.74

TABLE 7 | Results of five-fold cross-validation of the proposal ensemble method on the Kermay Pneuonia dataset.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
1	98.63	99.31	98.85	99.08
2	98.81	99.31	99.08	99.19
3	98.13	98.18	99.31	98.74
4	98.47	99.08	98.85	98.96
5	98.46	99.30	98.62	98.96
Avg $\pm$ std. dev.	98.50 $\pm$ 0.32	99.03 $\pm$ 0.49	98.80 $\pm$ 0.26	98.99 $\pm$ 0.17

Notes: 95% CI = (97.08%, 99.13%) with accuracy = 98.50%.

Abbreviations: Avg: average; Std. Dev.: standard deviation.

recall demonstrates reliable detection across all target classes. The resulting F1-score, the highest among all methods, signifies superior balance and robustness.

4.4 | Cross-Dataset Testing and Cross-Validation

The RSNA Pneumonia Detection Challenge dataset was utilised to provide an additional evaluation setting, and the performance metrics reported in Table 6 were used to compare different model configurations. Among the CNN-based ensemble models, the combination of InceptionResNetV2, MobileNetV2, and VGG16 achieved the best performance, with an accuracy of 78% and a precision of 82%. However, these results remain lower than those obtained by the proposed Pneu-ConvViT model, which achieved an accuracy of 84.45% and a precision of 85.05% on the same dataset (Table 6).

While these findings indicate that the hybrid CNN-ViT architecture maintains a performance advantage in this experimental setting, it is important to note that evaluation on a single external dataset is not sufficient to fully establish generalisation capability. Therefore, external validation on additional independent CXR datasets is strongly recommended to support more robust claims regarding generalisation and clinical applicability. Such validation would help ensure that the proposed model performs consistently across different data distributions, imaging conditions, and patient populations, thereby strengthening its potential for real-world deployment.

The RSNA dataset [44] was utilised for training, and the performance metrics displayed in Table 6 were used to assess each

model's performance and determine which ensemble model performed the best. With an accuracy of 78% and a precision of 82%, the combination of InceptionResNetV2, MobilenetV2, and VGG16 produced the best results in the CNN model ensembles. Nonetheless, these figures remain below the 84.45% accuracy and 85.05% precision of the Pneu-ConvViT model.

In this study, the performance of the suggested ensemble model was robustly assessed using a five-fold cross-validation scheme. Table 7 for the Kermay dataset [22] and Table 8 for the RSNA challenge dataset [44] contain the findings for each fold as well as the average and standard deviation values over the five folds. For the Kermay dataset, the model achieved an average accuracy of $98.50\% \pm 0.32\%$, with a 95% confidence interval of [97.08%, 99.13%], demonstrating highly stable and consistent performance. In contrast, evaluation on the RSNA dataset resulted in lower but reliable performance, reflecting the increased variability and complexity of real-world clinical data. These results confirm that the proposed hybrid CNN-ViT framework not only achieves high accuracy on benchmark datasets but also maintains robustness and generalisation across different data distributions.

In summary, the results clearly show that:

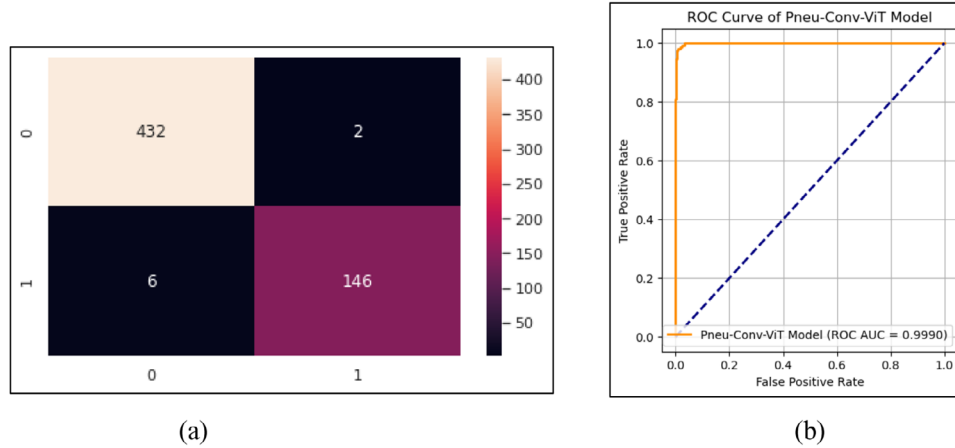
- Model diversity improves ensemble performance, especially when mixing lightweight and deep CNN architectures.
- Transformers alone are insufficient for this task but become highly effective when integrated with CNN-based models.
- The hybrid CNN-ViT framework achieves highly competitive performance by effectively capturing both local and global representations.

TABLE 8 | Results of five-fold cross-validation of the proposal ensemble method on the RSNA Pneumonia Detection Challenge dataset [44].

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
1	83	84.55	84.54	84.54
2	83	82.2	88.18	85.09
3	82	81.82	90	85.71
4	85.5	91.3	84.68	87.87
5	89	85.36	96.33	90.51
Avg $\pm$ Std. Dev.	84.50 $\pm$ 2.83	85.05 $\pm$ 3.81	88.75 $\pm$ 4.84	86.74 $\pm$ 2.46

Notes: 95% CI = (81.40%, 87.09%) with Accuracy = 84.50%.

Abbreviations: Avg: average; Std. Dev.: standard deviation.

**FIGURE 3** | The Pneu-ConViT in training by the Kermany dataset [22]: (a) confusion matrix with heatmap and (b) the ROC curve of the hybrid vision transformer model.

- Although the proposed Pneu-ConViT framework achieved 99.06% accuracy on the Kermany dataset, the accuracy decreased to 84.50% on the external RSNA dataset, indicating the presence of domain shift. This performance degradation can be attributed to differences in patient demographics, age distributions, imaging protocols, disease prevalence, image quality, annotation strategies, and acquisition equipment between the two datasets. In particular, the Kermany dataset consists primarily of paediatric CXRs, whereas the RSNA dataset contains a more heterogeneous patient population. Despite this reduction, the proposed framework maintained robust performance on the external dataset, demonstrating reasonable generalisation capability.

Overall, the proposed Pneu-ConViT framework establishes a highly competitive benchmark for pneumonia classification and highlights the effectiveness of feature-level fusion strategies for complex medical image analysis tasks.

4.5 | Confusion Matrix and ROC Analysis

In the Kermany Pneumonia Dataset, only 1.5% of pneumonia cases were misclassified as normal, primarily mild viral infections with low opacity contrast and subtle radiographic patterns. These results highlight the model's strong discriminative ability and its

robustness in detecting pneumonia cases, even under challenging low-intensity visual conditions. The confusion matrices also revealed that the hybrid model yielded higher true-positive rates in borderline cases, which radiologists identified as diagnostically challenging. This suggests that integrating ViT's attention mechanism with CNN's spatial feature extraction enhances the model's ability to discriminate subtle pneumonia patterns.

Furthermore, the ROC analysis demonstrated superior classification performance, achieving an ROC-AUC of 0.9990, a sensitivity of 99.53%, and a specificity of 96.05% on the Kermany dataset, indicating excellent balance between precision and recall as well as strong discriminative capability across clinical decision thresholds. Figure 3 displays the confusion matrices for the Kermany dataset.

4.6 | Visualisation Using Grad-CAM and Attention Maps

To interpret the decision-making process of the hybrid ViT-CNN ensemble for pneumonia classification on CXR images, we employed a combination of Grad-CAM for CNN backbones and Attention Rollout for the ViT-B/16 branch. These visualisation techniques allow us to identify the most salient regions in the input image that contributed to the model's prediction, providing

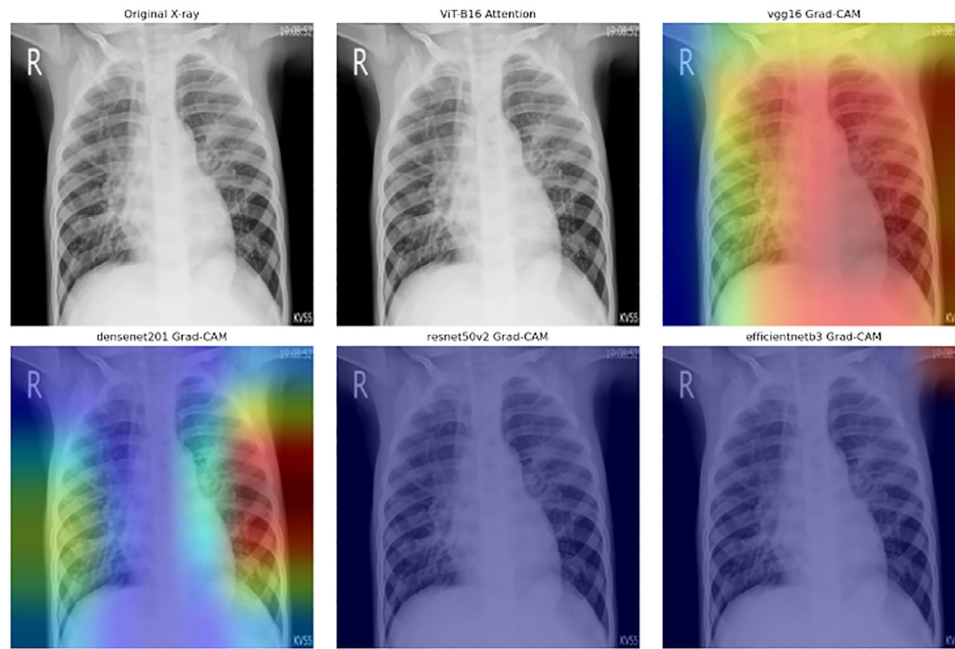


FIGURE 4 | Visualisation of model attention for pneumonia classification on chest X-rays using the hybrid Vision Transformer–CNN ensemble. The 2×3 grid shows (from left to right, top to bottom) original X-ray image; ViT-B/16 attention map generated via attention rollout; VGG16 Grad-CAM heatmap; DenseNet201 Grad-CAM heatmap; ResNet50V2 Grad-CAM heatmap; and EfficientNetB3 Grad-CAM heatmap.

insights into both the localised and contextual features leveraged by the ensemble.

Specifically, for the hybrid ViT–CNN ensemble, we generated visual explanations in three complementary ways:

- **Grad-CAM for CNN Branches:** Each CNN backbone in the hybrid model – VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 – produces a Grad-CAM heatmap by computing the gradient of the predicted class score with respect to the feature maps of the final convolutional layer. The resulting heatmaps were normalised and upsampled to match the original input resolution (224×224 pixels) and overlaid on the corresponding CXR images. These visualisations highlight regions that were most influential in driving the CNN predictions for pneumonia detection, as shown in Figure 4.
- **ViT-B/16 Attention Rollout:** As the ViT-B/16 branch produces a sequence embedding without spatial dimensions, we utilised the attention rollout method to propagate attention scores from the classification token through all transformer blocks. This procedure generates a 2D attention map that reflects the regions in the input image receiving the highest attention. The attention map was normalised and overlaid on the original X-ray image, providing an interpretable visualisation of the ViT's focus, as shown in Figure 5.
- **Fusion Hybrid Heatmap:** To provide a holistic visualisation of the hybrid ensemble, the Grad-CAM heatmaps from all CNN backbones were combined with the ViT attention map by averaging after normalisation. The resulting fusion heatmap overlay illustrates the consensus regions emphasised by the entire hybrid model. This approach captures both localised features detected by CNNs and the broader contextual patterns attended by the ViT, yielding a comprehensive representation of the model's reasoning process for pneumonia classification, as shown in Figure 5.

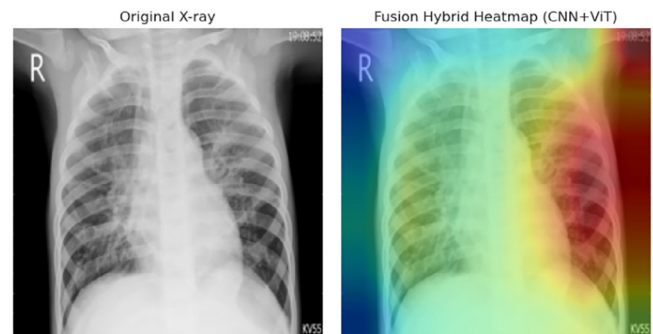


FIGURE 5 | Fusion visualisation of the hybrid ensemble combining four CNN backbones and the ViT-B/16 branch. The fusion heatmap is obtained by averaging the normalised Grad-CAM maps from VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 with the ViT-B/16 attention map.

sensation of the model's reasoning process for pneumonia classification, as shown in Figure 5.

The hybrid ensemble exhibits the following:

- Broader and anatomically aligned attention over the lung fields,
- Reduced focus on irrelevant artefacts (ribs, diaphragm), and
- Better alignment with radiologist-annotated pneumonia regions.

While CNNs alone tended to emphasise high-contrast areas, the ViT attention contributed context-aware localisation, capturing diffuse patterns characteristic of viral pneumonia. This

TABLE 9 | Comparison of computational complexity and training time for different models.

Models	Params (M)	FLOPs (G)	Memory (MB)	Training time (s)
MobileNetV2	2.42	0.30	9.68	8245
VGG16	14.78	15.50	59.12	20,145
InceptionResNetV2	54.53	13.20	218.12	21,496
DenseNet201	18.57	4.30	74.28	20,707
ResNet50V2	23.83	6.90	95.32	15,515
InceptionResNetV2 & VGG16	69.62	28.70	278.48	35,200
MobileNetV2 & ResNet50V2	26.72	7.20	106.88	15,757
InceptionResNetV2 & ResNet50V2	78.87	20.10	315.48	28,875
DenseNet169 & MobileNetV2	15.70	3.70	62.80	18,322
DenseNet201, VGG16, ResNet50V2	57.80	26.70	231.20	26,800
InceptionResNetV2, DenseNet201, ResNet50V2	97.69	24.40	390.76	36,886
DenseNet201, MobileNetV2, ResNet50V2	45.54	11.50	182.16	27,264
InceptionResNetV2, MobileNetV2, VGG16	72.21	29.00	288.84	36,893
InceptionResNetV2, DenseNet201, VGG16	88.44	33.00	353.76	26,012
Pneu-ConViT without CNN Ensemble (ViT-B/16 Only)	86.36	35.24	1373.51	77,404
Pneu-ConViT without ViT (CNN Ensemble Only)	70.66	48.23	1282.02	61,531
Pneu-ConViT	156.85	83.48	2634.28	90,354

interpretability reinforces the clinical credibility of the proposed hybrid framework.

4.7 | Computational Complexity and Memory Analysis

The computational complexity of the proposed models was evaluated using three key metrics: the number of trainable parameters, floating point operations (FLOPs), and memory consumption. These metrics provide a comprehensive understanding of both computational cost and deployment feasibility in clinical environments.

Table 9 summarises the complexity of all evaluated architectures, including individual CNNs, transformer-based models, ensemble configurations, and the proposed hybrid framework. The computational complexity of the evaluated models is analysed in terms of parameters, FLOPs, and memory usage. Lightweight architectures such as MobileNetV2 require minimal resources, while deeper CNNs (e.g., VGG16 and InceptionResNetV2) exhibit significantly higher computational costs.

The ViT-B/16 introduces global context modelling with approximately 17.60 GFLOPs and 86 M parameters (~344 MB), offering stronger representation capability at increased cost. Ensemble models further increase complexity, as their total FLOPs and parameters scale with the number of backbone networks. For example, the combination of VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 requires approximately 28.50 GFLOPs and 68.80 M parameters. The proposed Pneu-ConViT model has the highest complexity, with 83.48 GFLOPs and 156.85 M parameters (~627 MB), due to the integration of multiple CNN backbones and

a transformer module. This design enables the model to effectively capture both local spatial features and global contextual dependencies, leading to superior classification performance. Although computationally intensive, such a trade-off is justified in medical imaging applications, where diagnostic accuracy and reliability are more critical than efficiency. Moreover, the computational cost can be reduced in practical deployment through optimisation strategies such as pruning, distillation, or backbone selection. Overall, the results highlight a trade-off between computational efficiency and predictive performance. While lightweight models are suitable for resource-constrained environments, the proposed hybrid framework is more appropriate for high-performance systems. All FLOPs are computed with an input resolution of 224×224 and represent the cost of a single forward pass.

4.8 | Comparative Study With Recent Models in Chest X-Ray Images

The Pneu-ConViT framework, a hybrid ViT-CNN ensemble specifically developed for pneumonia classification on the Germany Pneumonia dataset, integrates five pre-trained backbones – ViT-B/16, VGG16, DenseNet201, ResNet50V2, and EfficientNetB3 – thereby combining the global self-attention capabilities of ViT with the hierarchical local feature extraction of multiple CNNs. Achieving 98.63% accuracy, 99.90% AUC, and 98.96% F1-score, Pneu-ConViT demonstrates superior performance compared with several state-of-the-art approaches.

The proposed Pneu-ConViT framework demonstrates superior performance compared with several state-of-the-art approaches, as summarised in Table 10. It is important to note that, in this

TABLE 10 | Comparison with recent ensemble and hybrid models for the binary classification task in the Kaggle pneumonia dataset.

Author	Backbone	Accuracy (%)
Mabrouk et al. [47]	ViT + DenseNet169, MobileNetV2	93.91
Saber et al. [48]	TransUNet	93.75
Chaudhary and Saroj [49]	ConvMixer Ensemble	98.0
Ukwuoma et al. [50]	Hybrid ViT and CNNs	97.2
Roy et al. [51]	FA-Net	97.15
Al-Hejri et al. [52]	ViT-B16 + DenseNet201, Xception, and InceptionResNetV2	97.26
Dash et al. [53]	GAN + ViT	97.3
Singh et al. [54]	Custom ViT	97.61
Mustapha et al. [55]	Modified swin transformer	98.72
Hadhoud et al. [56]	ViT-B16 + ResNet50	98.8
Pneu-ConViT	ViT + CNNs	98.63

study, the Kermay CXR Dataset is reformulated as a binary classification task, where the original labels (normal, bacterial pneumonia, and viral pneumonia) are consolidated into two categories: NORMAL and PNEUMONIA. Specifically, both bacterial and viral pneumonia cases are merged into a single pneumonia class to align with common clinical screening scenarios, where the primary objective is to distinguish healthy subjects from those requiring further diagnostic attention. This binary setting simplifies the classification problem while preserving clinical relevance, and it enables more direct comparison with prior studies that adopt a similar protocol. Under this configuration, the proposed framework achieves highly competitive results, highlighting its effectiveness in capturing both local pathological features and global contextual patterns for robust pneumonia detection.

Mabrouk et al. [47] utilise an ensemble of deep CNN architectures to achieve high accuracy; however, the approach lacks the capability to capture long-range dependencies, which may limit its effectiveness in detecting subtle pathological patterns. Saber et al. [48] employ TransUNet and achieve 93.75% accuracy, indicating the potential of transformer-based architectures, although the method is not specifically optimised for hybrid CNN-ViT feature fusion in pneumonia classification. Chaudhary and Saroj [49] propose a ViT-VGG19 ensemble that attains 98.0% accuracy, but the method does not fully optimise the contribution of individual models within the ensemble. Ukwuoma et al. [50] introduce a hybrid explainable transformer-based ensemble combining DenseNet201, VGG16, and GoogleNet, achieving 98.19% accuracy; nevertheless, the framework lacks domain-specific optimisation tailored to pulmonary imaging characteristics. Roy et al. [51] present FA-Net, which integrates DenseNet121 and ResNet50 with fuzzy contextual spatial self-attention, achieving 97.15% accuracy, although it is limited to dual backbones and may not fully exploit diverse feature representations. Al-Hejri et al. [52] further extend hybrid modelling by combining ViT-B16 with multiple CNN backbones, including DenseNet201, Xception, and InceptionResNetV2, achieving an accuracy of 97.26%. While this ensemble improves representation diversity, it relies primarily on architectural aggregation without incorporating data-centric enhancement or systematic optimisation strategies. Similarly,

Dash et al. [53] integrate GANs with ViTs to enhance data diversity, achieving 97.3% accuracy. Although this approach demonstrates the effectiveness of GAN-based augmentation, it does not fully explore hybrid CNN-ViT feature fusion or adaptive weighting mechanisms. Singh et al. [54] develop a customised ViT and achieve 97.61% accuracy, showing the effectiveness of transformer-based global representation learning, but without explicitly integrating complementary CNN-based local features. Mustapha et al. [55] employ a modified Swin Transformer and report 98.72% accuracy, demonstrating the advantage of hierarchical attention mechanisms; however, the approach remains primarily transformer-centric and does not fully exploit multi-backbone CNN-ViT fusion.

In comparison, the proposed method achieves an accuracy of 99.06%, surpassing the 98.8% reported by Hadhoud et al. [56], one of the strongest hybrid CNN-ViT baselines. This performance gain can be attributed to the synergistic integration of contrast-enhancing preprocessing, GAN-based data augmentation, and a DNN-based learnable fusion strategy that effectively combines CNN-derived local features with ViT-based global contextual representations. While slight variations in experimental protocols may exist across studies, the results demonstrate that the proposed framework provides a competitive and robust solution for pneumonia classification under the binary classification setting.

Overall, while recent CNN-Transformer and transformer-based approaches have demonstrated promising results, the proposed Pneu-ConViT framework achieves highly competitive performance for pneumonia classification. The effectiveness of the framework stems from the integration of complementary local and global feature representations through a learnable DNN-based feature fusion module, combined with CLAHE-enhanced preprocessing and GAN-based data augmentation. Although the individual components employed in this study have been previously reported, their integration within a unified end-to-end framework distinguishes Pneu-ConViT from many existing approaches. By combining multi-backbone CNN feature extraction, ViT-based contextual modelling, feature-level fusion, and interpretability analysis, the proposed framework provides a robust and clinically relevant solution for CXR classification

while maintaining strong generalisation capability across different datasets.

4.9 | Ablation Study

To assess each module's contribution, we performed controlled ablations:

- Without ViT: A CNN-only ensemble reduced AUC by 0.3% and accuracy by 0.7%.
- Without EfficientNetB3: Loss of 0.3% in accuracy, confirming its efficiency scaling benefit.
- Without DNN-based learnable fusion: Equal averaging of Sigmoid outputs decreased F1-score from 99.08% to 93.86%, highlighting the importance of learned weights.
- Without Preprocessing Enhancements: Removing CLAHE caused the accuracy and recall to drop by $\sim 0.2\%$ and $\sim 1\%$.

The contribution of the CLAHE-based preprocessing step was evaluated through controlled ablation experiments. Removing CLAHE from the pipeline results in a consistent decline in performance across all evaluation metrics, particularly in recall and overall accuracy. This degradation can be attributed to the reduced visibility of subtle pulmonary patterns, such as low-contrast opacities and interstitial textures, which are critical for accurate pneumonia detection. By enhancing local contrast while preventing noise amplification, CLAHE facilitates the extraction of more discriminative features, thereby improving the effectiveness of both CNN-based local feature learning and ViT-based global context modelling. These findings confirm that CLAHE plays a crucial role in strengthening feature representation and contributes significantly to the robustness of the proposed framework.

The impact of the fusion mechanism was further analysed by comparing a DNN-based fusion head with a simple concatenation strategy. Experimental results show that naive concatenation without subsequent learnable layers fails to effectively exploit the complementary information across model outputs, leading to suboptimal performance. In contrast, the DNN fusion head enables the learning of non-linear interactions between heterogeneous feature representations, resulting in improved decision boundaries and enhanced classification performance. Quantitatively, replacing the DNN fusion head with simple concatenation leads to a noticeable degradation in F1-score, confirming the importance of incorporating learnable transformation layers after feature aggregation.

While weighted fusion offers robustness and interpretability through linear aggregation of model outputs, it lacks the flexibility required to model complex cross-representation interactions. To address this limitation, we adopt a learned attention-based fusion strategy, realised through feature concatenation followed by a DNN-based fusion module. This design enables the network to learn adaptive, non-linear feature relationships and to dynamically reweight contributions from different model branches in a data-dependent manner. As a result, the proposed fusion approach enhances the integration of complementary local and

global representations and leads to improved classification performance. The observed gains, particularly in reducing misclassification of visually ambiguous cases, highlight the advantage of learned fusion mechanisms over traditional weighted fusion. These results underscore the synergistic effect of ViT inclusion and adaptive fusion weighting.

4.10 | Analytical Discussion

The hybrid ensemble's superior performance stems from three complementary mechanisms:

- Hierarchical Feature Integration: CNNs encode fine-grained local textures, while ViT captures holistic context; their concatenation provides a more complete representation of pulmonary structures.
- Cross-Model Complementarity: Diverse inductive biases among backbones reduce correlated errors, improving ensemble stability.
- Improved Generalisation: Training exclusively on the Kermanshah Pneumonia dataset, combined with the proposed DNN-based learnable fusion strategy, enhances the model's ability to generalise across diverse patient presentations and varying radiographic characteristics. By leveraging feature concatenation followed by a dedicated fusion network, the model effectively captures non-linear and complementary interactions between heterogeneous backbone representations. This adaptive integration mechanism enables more robust feature discrimination and improves generalisation performance, thereby ensuring reliable behaviour in clinically relevant scenarios.

Moreover, the attention visualisation confirms that the hybrid model prioritises clinically relevant lung zones, an essential aspect for AI-assisted radiology adoption. These characteristics affirm that transformer-based attention can complement convolutional inductive priors even in limited-data scenarios, provided ensemble regularisation is carefully designed.

4.11 | Limitations

Despite the promising outcomes, several challenges remain:

- The hybrid model's computational cost is higher than single CNN inference, suggesting a need for model-compression techniques such as knowledge distillation.
- The current training relies on supervised labels; future work will explore self-supervised pretraining to leverage large unlabelled CXR repositories.

Extending to multi-class pulmonary diseases (e.g., tuberculosis, COVID-19, and fibrosis) could further demonstrate general utility.

5 | Conclusion

This study proposed a comprehensive hybrid ViT–CNN ensemble framework for automated pneumonia classification using CXR images from the Kermanshah dataset. The model was designed

to combine the complementary strengths of transformer-based global attention and convolution-based local feature extraction to achieve higher diagnostic accuracy and interpretability. Specifically, the Pneu-ConViT framework integrates the ViT-B/16 with multiple convolutional backbones – VGG16, ResNet50V2, DenseNet201, and EfficientNetB3 – to extract rich hierarchical and contextual representations from radiographic data. The multi-scale features from these networks were concatenated and passed through a DNN fusion head, consisting of fully connected layers with ReLU activations and dropout regularisation, followed by a final sigmoid activation function to produce binary probabilistic predictions. This DNN fusion layer serves as the classification head, effectively learning cross-model dependencies and reducing redundancy among the diverse feature sets. The Pneu-ConViT framework was trained and evaluated on the publicly available Kermany dataset, which is widely recognised as a benchmark for pneumonia classification. Quantitative results demonstrate that the hybrid ensemble model achieved 99.06% accuracy, 0.9863 precision, 0.98 recall, 0.9909 F1-score, and 0.9990 AUC on the Kermany dataset, significantly outperforming individual CNN or ViT baselines. The superior performance confirms the effectiveness of the ensemble-based fusion approach in leveraging both global and local discriminative information.

Furthermore, Grad-CAM visualisation analyses illustrated that the hybrid model accurately highlighted pneumonia-infected regions, focusing on relevant lung areas and lesion boundaries, thereby improving the transparency and clinical interpretability of the predictions. The results also indicate that the model's decision process aligns well with radiological patterns recognised by human experts, validating its potential as a reliable decision-support tool for medical practitioners. Overall, the integration of attention-based transformers and CNNs provides a robust and generalisable approach for CAD in medical imaging. This hybrid ensemble model offers not only improved diagnostic accuracy but also enhanced resilience against noise, variability in imaging conditions, and dataset imbalance.

Future research will focus on optimising the architecture through self-supervised pretraining and contrastive learning to further improve feature generalisation; developing lightweight hybrid variants for real-time clinical and mobile deployment; and extending the binary classification framework to multi-label detection of multiple thoracic pathologies, such as fibrosis, oedema, and effusion. The promising results of this work highlight the growing potential of hybrid DL models to transform radiological analysis, providing faster, more accurate, and interpretable diagnostic support in healthcare applications.

Author Contributions

Ngoc Ha Pham: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Giang Son Tran:** supervision, writing – review and editing. **Antoine Doucet:** supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used in this study are publicly available: 1. Kermany Chest X-ray dataset (Mendeley Data). Kermany, D., Zhang, K., Goldbaum, M.: 'Labeled optical coherence tomography (OCT) and chest X-ray images for classification'. *Mendeley Data*, 2018. 2. RSNA Pneumonia Detection Challenge dataset. Shih, G., Wu, C. C., Halabi, S. S., Kohli, M. D., Prevedello, L. M., Cook, T. S., Sharma, A., Amorosa, J. K., Arteaga, V., Galperin-Aizenberg, M., et al.: Augmenting the National Institutes of Health Chest Radiograph Dataset with Expert Annotations of Possible Pneumonia. *Radiology: Artificial Intelligence*, 2019, 1(1), e180041. Processed data and code can be made available upon reasonable request to the corresponding author.

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